

Product Requirements Document

Enterprise AI Investment Analyst Platform (EIAP)

Owner: [Your Name] | **Role:** AI Product Manager
(Portfolio Project)

Status: V1 — Design Complete, Build in Progress

Stack: No-Code Visual AI Orchestration (Flowise AI /
Dify.ai)

1. Product Vision

Indian equity research analysts spend 4–6 hours per company reading 150–300 page annual reports (10-K equivalents, i.e., Form AR/Integrated Annual Reports filed with BSE/NSE) to extract three things: **financial ratios buried in schedules, forward-looking management commentary, and litigation/related-party risk disclosures**. This is high-value cognitive work trapped in low-value manual retrieval.

Vision Statement: *Give every equity analyst in India a research associate that has read every annual report in the coverage universe, never forgets a footnote, and never asserts a number it cannot cite.*

EIAP is not a "give me a stock tip" chatbot. It is a **grounded retrieval and synthesis tool** that sits between raw filings and the analyst's own judgment — the AI retrieves and cites, the human still decides.

2. User Persona

Primary Persona: "Priya" — Junior Equity Research Analyst, Mumbai-based brokerage/AMC

Attribute	Detail
Experience	1–3 years, covers 15–20 stocks across a sector (e.g., Auto Ancillaries)
Core Task	Reads annual reports, builds models, writes initiation/update notes
Pain Point	Retrieval fatigue — knows the answer exists in the AR but not on which of 280 pages

Attribute	Detail
Tooling today	Screener.in, Excel, PDF + Ctrl+F, Bloomberg terminal (if senior enough)
Trust Bar	Extremely high — a single hallucinated number in a client-facing note is a career risk
Regulatory Context	Works under a SEBI-registered entity; personally may or may not be a SEBI Research Analyst (RA)

Secondary Persona: Portfolio Manager who spot-checks the analyst's claims before a client call — needs **fast verification**, not fast generation.

3. Problem Statement

"I don't need an AI that *guesses* what a company's contingent liabilities are. I need an AI that finds the exact paragraph, in under 10 seconds, with the page number, so I can quote it in my note without re-reading 40 pages."

The product does not compete on creativity. It competes on **retrieval precision + citation integrity + speed**, in that priority order.

4. Functional Specifications

4.1 Financial RAG Pipeline

Objective: Answer natural-language questions grounded strictly in the ingested annual report corpus, with page-level citations.

Component	Spec
Ingestion	PDF Loader → per-company, per-year document namespace (metadata: <code>ticker</code> , <code>fy_year</code> , <code>doc_type</code>)
Chunking	800 tokens / 100 token overlap for narrative sections (MD&A, Chairman's letter); table-aware chunking (whole-table-as-chunk, no mid-table splits) for financial statements — tables must never be split across chunk boundaries, since a split balance sheet row is worse than no retrieval at all
Metadata Filtering	Every retrieval call is pre-filtered by ticker + fiscal year before semantic search runs — this is non-negotiable. Without it, the system can retrieve "Revenue: ₹4,200 Cr" from the wrong company's FY23 report and present

Component	Spec
	it as the queried company's FY24 number. This is the single highest-severity failure mode in financial RAG.
Retrieval	Hybrid: dense vector similarity (top-k=8) + BM25 keyword match (top-k=4) → re-ranked (cross-encoder) → top 5 passed to LLM context
Citation	Every generated claim is mapped back to {source_doc, page_number, chunk_id} and rendered as an inline footnote in the UI — no citation, no claim is displayed

PM Trade-off logged: Hybrid retrieval (dense + BM25) costs ~2x the vector-search-only latency but is mandatory here — financial documents are full of exact-match critical terms (ticker symbols, "contingent liability," specific rupee figures) that pure semantic embeddings systematically under-retrieve. I am trading latency for precision because the cost of a *missed* retrieval (silent gap) is higher than the cost of a *slow* retrieval in this persona's workflow.

4.2 Live Stock Price / Market Data Tool

Objective: Answer real-time questions ("What's the current P/E of this stock?") that a static, ingested-in-January annual report cannot answer.

Component	Spec
Trigger	Router/Agent node classifies query intent: historical/document vs. live/market vs. hybrid
Tool	REST API call node (e.g., NSE/BSE data API or a licensed vendor like Screener/Tickertape API) returning LTP, P/E, 52-week range, market cap
Freshness Guardrail	Every live-data response is timestamped in the UI: "As of 3:41 PM IST" — the system must never let a live price sit silently next to a static FY23 filing number without disambiguating which is which
Hybrid case	"Is this stock overvalued vs. its 5-year average P/E from the AR?" → Agent fires both tools, then the LLM is prompted to explicitly separate "live data" vs "filed data" in its answer, not blend them into one unlabeled number

4.3 Guardrails Against Financial Data Hallucination

This is the section that separates a toy demo from an enterprise-credible system.

1. **Retrieval-Grounded Generation Only** — System prompt enforces: *"You may only state a figure if it appears verbatim in the retrieved context below. If the answer is not in the context, respond: 'This information is not available in the ingested filing for [ticker, FY].' Do not estimate, infer, or use general financial knowledge to fill gaps."*
2. **Numeric Verification Layer** — A lightweight post-generation check (regex/pattern match) confirms every numeric token in the LLM's output exists verbatim in the retrieved chunks before the response is shown to the user. Mismatches trigger a fallback "unable to verify" response instead of silently displaying an unverified number.
3. **No-Advice Guardrail (see Section 5)** — Hard-coded refusal pattern for any query classified as investment advice ("Should I buy X?").
4. **Confidence Thresholding** — If the top retrieved chunk's similarity score falls below a set threshold (e.g., 0.75 cosine), the system declines to answer rather than generating from a weak match ("low-

confidence retrieval — please rephrase or the information may not be in this filing").

5. **Human-in-the-loop Flagging** — Every response carries a "Flag as incorrect" button feeding directly into the Data Flywheel (Section 6.2).

5. Guardrails & Compliance (SEBI Context)

India-specific compliance framing — this is the section that shows regulatory maturity, not just AI maturity.

Rule	Enforcement
No Buy/Sell/Hold recommendations — Under SEBI (Research Analysts) Regulations, 2014, only a SEBI-registered RA/RIA may issue investment advice or recommendations. This tool is a research retrieval assistant , not an advisor.	Intent classifier detects advice-seeking queries ("should I invest," "is this a good buy") and returns a fixed disclaimer response instead of routing to the LLM at all — refusal happens before generation, not after.

Rule	Enforcement
No price targets or return projections	System prompt explicitly bans generating forward price/return numbers not directly quoted from a licensed research source in the corpus.
Mandatory disclaimer on every session	Persistent UI banner: <i>"This tool retrieves and summarizes information from public filings. It is not investment advice and is not a substitute for a SEBI-registered advisor."</i>
Data provenance only from official filings	Corpus is restricted to BSE/NSE/company-filed documents — no scraped forum content, no unverified social media sentiment, to avoid propagating misinformation as "cited fact."
Audit log	Every query + response + citations is logged with a timestamp for

Rule	Enforcement
	traceability, mirroring how a compliance desk would need to review analyst-AI interactions.

6. Success Metrics & Data Flywheel

6.1 Target Metrics (V1 Launch Bar)

Metric	Target	Why this number
P95 Latency (document RAG query)	< 6 seconds	Analyst attention span for a "quick lookup" task; beyond this they revert to Ctrl+F
P95 Latency (live price query)	< 2 seconds	Users expect near-instant for market data — matches Bloomberg/Screener expectations
Faithfulness (Ragas)	> 0.90	Near-zero tolerance for unfaithful claims in a financial context — this is the single

Metric	Target	Why this number
		most important quality metric for this product
Context Precision (Ragas)	> 0.80	Ensures retrieved chunks are actually relevant, not just topically adjacent
Context Recall (Ragas)	> 0.85	Ensures the <i>right</i> chunk wasn't missed by the retriever
Cost per query	< ₹1.50 (~\$0.018)	At 500 queries/day this caps monthly LLM spend at a defensible ~₹22,500 for a portfolio-stage/small-team product
Citation Rate	100% of factual claims	Non-negotiable — this is the trust mechanism of the entire product

6.2 Data Flywheel

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User Query → RAG Response + Citation
      ↓
"Correct" / "Flag Incorrect" feedback (1-
click)
      ↓
Flagged responses → weekly review queue
      ↓
Root-cause tagged: [Retrieval Miss |
Chunking Error | Hallucination | Wrong Doc
Filter]
      ↓
Feeds back into:
- Chunking strategy tuning (if Chunking
Error)
- Embedding model / hybrid weight tuning
(if Retrieval Miss)
- Prompt/guardrail tightening (if
Hallucination)
- Metadata tagging fix (if Wrong Doc
Filter)
      ↓
Re-run offline Ragas eval suite against a
frozen 50-question golden test set
before deploying the fix → prevents
regression
```

PM insight embedded here: the flywheel is not "collect data to fine-tune a model" — for a RAG system, the highest-leverage flywheel loop is **retrieval and chunking improvement**, not model fine-tuning. This is a deliberate architectural point of view.

7. Out of Scope (V1)

- Portfolio-level recommendation or allocation advice
 - Real-time trading execution / order placement
 - Sentiment analysis from unverified social sources
 - Multi-language support (V1 = English-language filings only)
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8. Key Risks

Risk	Mitigation
Hallucinated financial figures reach a client note	Numeric verification layer + mandatory citation display
Cross-company data contamination in retrieval	Hard metadata pre-filtering by ticker+FY before vector search
Regulatory overreach (tool perceived as giving advice)	Intent-classification refusal layer, persistent disclaimer, no recommendation language in system prompt

Risk	Mitigation
Stale live-data presented as current	Explicit timestamping on every live API response